ASSIGNMENT 1 – CO1 | TASK 10

Compare Expectations with Execution

Scenario: Namma Metro Smart-Card Fare System

Objective / explanation

This program compares integer-paise and double-rupee accumulation for a large number of journeys. The input example assumes each journey costs Rs 30.

C++ source code

```
#include &lt;iomanip&gt;
#include &lt;iostream&gt;
using namespace std;

int main() {
    long long journeys = 3000000LL;
    long long integerPaise = journeys * 3000LL;
    double doubleRupees = journeys * 30.0;
    cout &lt;&lt; "Integer paise total: " &lt;&lt; integerPaise &lt;&lt; '\n';
    cout &lt;&lt; fixed &lt;&lt; setprecision(2);
    cout &lt;&lt; "Double rupees total: " &lt;&lt; doubleRupees &lt;&lt; '\n';
    cout &lt;&lt; "Integer total in rupees: " &lt;&lt; integerPaise / 100 &lt;&lt; '\n';
}
```

Compilation

```
g++ -std=c++17 -Wall -Wextra -pedantic main.cpp -o program
```

Actual captured program output

```
Integer paise total: 9000000000
Double rupees total: 90000000.00
Integer total in rupees: 90000000
```

Output screenshot

Analysis / complexity

A 32-bit signed int cannot hold 9,000,000,000 paise. long long can. Double holds this magnitude, but decimal fractions are not always represented exactly. Additions are O(1) for each type.

Evidence note: Output shown is captured by executing this source in the generation environment. Recompile and run on your own system, and use your own screenshots if your instructor requires evidence from your computer.